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31 January 1989
SUPERSEDING
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15 March 1988

COMMERCIAL ITEM DESCRIPTION

CALIBRATION AND VERIFICATION SYSTEM

The General Services Administraton has authorized the use of this Commerical Item Description.

This Commercial Item Description covers a portable, battery operated, calibration and verification system for measuring to BRH standards the diagnostic X-Ray systems output. Consists of a non-invasive kVp measuring device, non-invasive exposure duration timer, and a generator milliampere second meter.

Salient characteristics:

The unit shall be a calibration/verification system to accurately measure, to BRH standards, the output of diagnostic x-ray systems. The calibration verification system shall measure kilovolt-peak (kVp), exposure duration in milisecond (ms) and/or second (s) and actual number of pulses for single phase x-ray generators, and x-ray generator millampere-second (mAs). The calibration/verification system shall accomplish the required kilovolt-peak and exposure duration measurements non-invasively. The milliampere-second measurement may be made invasive or non-invasive. The system shall include the instruments and respective connecting cables as follows:

a. Non-invasive kVp measuring device.
b. Non-invasive exposure duration timer.
c. Generator millampere-second measuring device.

Components included in the system shall consist of a maximum of three solid state devices and shall be packed in a single protective container. The outside dimensions of the container shall be 12-18" (L) X 10-16" (W) X 4-8" (H) and have a total system weight between 5 and 18 pounds.

The system shall have a service life, to include storage, of at least seven years (excludes batteries).

AMSC N/A FSC 6625

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The calibration and verification system shall have the following design features:

a. Subassemblies and chassis components which are removable without removing any other hard-wire subassembly or printed circuit board or component.

b. Permit adjustments without removing any component, printed circuit board, or module. The use of a circuit board/module extender is acceptable.

c. Printed circuit boards which are removable without the need to unsolder cables or wires.

The calibration and verification system shall have an operating temperature range of at least 15°C to 35°C, relative humidity to 90 percent. Storage temperatures shall be at least -35°C to 50°C, relative humidity to 90 percent.

The system shall be battery operated and utilize batteries that are currently available through Defense Supply System.

The system shall be capable of maintaining tolerance specifications for a period no less than 360 days. The system shall be capable of being field tested for verification of calibration utilizing the system instruments in a non-radiation environment.

The system shall be used in temporary or mobile shelters normally issued to medical TOE units, shall be packaged to withstand ground mobile transport over secondary and tributary roads and cross country terrain without adverse effects, and shall be compatible without additional packaging.

The High Voltage Measuring Device shall have operating capabilities as follows:

a. Measure kVp of an x-ray system without physically making electrical connections to the x-ray system in any manner.
b. Provide a signal(s) which can be displayed on an oscilloscope or digital multimeter. The signal must be proportional to true KV and independent of all other (MA or distance) electrical and radiation effects.
c. Provide an output signal for use with a timing device.
d. Be operational with single phase, three phase, half wave, full wave or self rectified x-ray system, and high frequency generators, dental and Cine x-ray generators.

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e. Frequency range of detection of the kVp meter shall range from direct current (DC), as produced by constant potential generators, to an upper limit 50 KHz, as produced by high frequency generators, and shall be capable of measuring one line cycle.
f. Shall have a range of 50-150 kVp, ± 2 percent of reading. Operating temperature range shall be 15° -35°C. Storage temperature range shall be -35°-50°c. Shall have an operating humidity range of 0-90 percent.

The time measuring device shall measure x-ray exposure times and shall have the following capabilities:

a. Pulse count range(s) of at least 0 to 9999 for single phase generators.
b. Exposure time range(s) of at least 0 to 9.999s ±0.1 percent +1 digit.

The Milliampere Second (mAs) Measuring Device shall measure the amount of current present in the secondary of the high tension transformer of the x-ray system during an exposure period.

DC current (mA) ranges of 20, 200, and 2,000 mA with an accuracy of +/- one percent and 2 digits, a response time of 1 second maximum and milliamp seconds (MAS) ranges of 200 MAS and 2,000 MAS with 3 on accuracy of +/- one percent and 2 digits MAS drift to be less than 0.25 percent of reading per minute. Operating temperature is between 5°C and 35°C and storage temperature of -17°C and 50°C, and relative humidity between 0 and 80 percent. Readout can be either integrated into unit or connected to a digital voltmeter.

Workmanship. The calibration and verification system shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One calibration and verification system, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 inspection level of S-2. The unit of product for examination purposes shall be one calibration and verification system.

DEFECTS.

Component missing.

Component damaged including (but not restricted to) cracks, checkings, breaks and deformed areas.

Exposed sharp edges or corners accessible to operatyor during normal use or cleaning.

Digital display not operable (if applicable).

Any switches not operable through their full range.

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DEFECTS. (continued)

Identification marking not complete, permanent, and legible.

Finish not free of chips, cracks, scratches, and other appearance defects.

Any component part not secured in place.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document:

6625-01-268-0383

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6625-0695

Location:
ICID/CALIBRATION/H17-31

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